Chapter 11

Profit and Loss

Profit and loss are the terms related to monetary transactions in trade and business. Whenever a purchased article is sold, then either profit is earned or loss is incurred.

Cost Price (CP): This is the price at which an article is purchased or manufactured.

Selling Price (SP): This is the price at which an article is sold.

Overhead Charges: Such charges are the extra expenditures on purchased goods apart from actual cost price. Such charges include freight charges, rent, salary of employees, repairing cost on purchased articles etc.
Note: If overhead charges are not specified in the question, then they are not considered

Profit or Gain (SP>CP): When an article is sold at a price more than its cost price, then profit is earned,

Loss (CP>SP): When an article is sold at a price lower than its cost price, then loss is incurred.

Ex. 1 A man buys an article for ₹300 and sells it for ₹900. Find profit/loss.

Sol. Here, SP $>$ CP.

$\therefore$ Profit is earned.

According to the formula,

$$\text{Profit} = \text{SP} - \text{CP} = 900 - 300 = ₹600$$

Ex. 2 Raman purchased a car for ₹5 lakh and sold it for ₹4 lakh. Find profit/loss in this transaction.

Sol. Here, SP $<$ CP.

$\therefore$ Loss is incurred in this case.

According to the formula,

$$\text{Loss} = \text{CP} - \text{SP} = (5-4) \text{ lakh} = | ₹1 \text{ lakh}$$

Table 11.1: Basic Formulas Related to Profit and Loss

1. Profit= SP−CP
2. Loss= CP−SP
3. $\text{Profit\%} = \frac{\text{Profit}}{\text{Cost Price}} \times 100 = \frac{P}{\text{CP}} \times 100$
4. $\text{Loss\%} = \frac{\text{Loss}}{\text{Cost Price}} \times 100 = \frac{L}{\text{CP}} \times 100$
5. $\text{SP} = \left(\frac{100 + \text{ Gain\%}}{100}\right) \times \text{CP}$
6. $\text{SP} = \left(\frac{100 - \text{ Loss\%}}{100}\right) \times \text{CP}$
7. $\text{CP} = \left(\frac{100}{100 + \text{ Gain\%}}\right) \times \text{SP}$
8. $\text{CP} = \left(\frac{100}{100 - \text{ Loss\%}}\right) \times \text{SP}$

Note

1. Profit and loss are always calculated on cost price unless otherwise stated in the question.
2. If an article is sold at a certain gain (say 45%), then SP = 145% of CP
3. If an article is sold at a certain loss (say 25%), then SP = 75% of CP

Ex. 3 A person buys a toy for ₹50 and sells it for ₹75. What will be his gain per cent?

Sol. Given,CP = ₹50 and SP = ₹75

Profit = SP -CP = 75 - 50 = ₹25

According to the formula,

$$\text{Gain\%} = \frac{\text{Profit}}{CP} \times 100 = \frac{25}{75} \times 100\% = 50\%$$

Ex. 4 A person buys a cycle for ₹450 but because of certain urgency, he sells it for ₹350. Find his loss per cent.

Sol. Given,CP = ₹450 and SP = ₹350

Loss = CP - SP = 450 - 350 = ₹100

According to the formula,

$$\text{Loss\%} = \frac{\text{Loss}}{CP} \times 100 = \frac{100}{450} \times 100\% = \frac{200}{9}\% = 22\frac{2}{9}\%$$

Ex. 5 Find the SP, when CP is ₹80 and gain is 20%.

Sol. Given, CP = ₹80 and Gain = 20%

$$\text{SP} = \left(\frac{100 + \text{ Gain\%}}{100}\right) \times CP$$

$$\text{SP} = \left(\frac{100+20}{100}\right) \times 80 = \frac{120}{100} \times 80 = 12 \times 8 = ₹\,96$$

Ex. 6 Find the SP when CP is ₹80 and loss is 20%.

Sol. Given, CP = ₹80 and Loss =20%

$$\text{SP} = \left(\frac{100 - \text{ Loss\%}}{100}\right) \times CP$$

$$\text{SP} = \left(\frac{100-20}{100}\right) \times 80 = \frac{80}{100} \times 80 = 8 \times 8 = ₹\,64$$

Ex. 7 Find the CP when SP is ₹40 and gain is 15%.

Sol. Given, SP = ₹40 and Gain = 15%

$$\text{CP} = \left(\frac{100}{100 + \text{ Gain\%}}\right) \times SP$$

$$\text{CP} = \left(\frac{100}{100+15}\right) \times 40 = \frac{100}{115} \times 40 = ₹\,34.78$$

Ex. 8 Find the CP when SP is ₹200 and loss is 35%.

Sol. Given, SP = ₹200 and Loss = 35%

$$\text{CP} = \left(\frac{100}{100 - \text{ Loss\%}}\right) \times SP$$

$$\text{CP} = \left(\frac{100}{100-35}\right) \times 200 = \frac{100}{65} \times 200 = ₹\,307.6$$

Ex. 9 A toy is bought for ₹150 and sold at a gain of 8%. Find its selling price.

Sol. Given, CP = ₹150 and Gain = 8%

$$\therefore \quad \text{Selling Price} = \left(\frac{100 + \text{ Gain\%}}{100}\right) \times CP = \left(\frac{100+8}{100}\right) \times 150$$

$$= \frac{108}{100} \times 150 = 54 \times 3 = ₹\,162$$

Ex. 10 A table is bought for ₹1500 and sold at a loss of 6%. Find its selling price.

Sol. Given, CP = ₹1500 and Loss = 6%

$$\therefore \quad \text{Selling Price} = \left(\frac{100 - \text{ Loss\%}}{100}\right) \times CP = \left(\frac{100-6}{100}\right) \times 1500$$

$$= \frac{94}{100} \times 1500 = ₹\ 1410$$

Ex. 11 By selling a watch for ₹1440, a man losses 10%. At what price should he sell it to gain 10%?

Sol. Given, SP = ₹1440 and Loss = 10%

$$\therefore \quad \text{Cost Price } = \left(\frac{100}{100 - \text{ Loss\%}}\right) \times SP = \left(\frac{100}{100 - 10}\right) \times 1440$$

$$= \frac{100}{90} \times 1440 = ₹\ 1600$$

Now, CP = ₹1600 and Gain = 10%

$$\therefore \quad \text{Required SP} = 1600 \times \frac{110}{100} = ₹1760$$

Ex. 12 Ravish lost 20% by selling a radio set for ₹3072. What per cent will he gain by selling it for ₹4080?

Sol. Given, SP = ₹3072 and loss = 20%

$$\therefore \quad CP = \frac{100}{80} \times 3072 = ₹3840$$

Now, CP = ₹3840 and SP = ₹4080

$$\text{Gain} = 4080 - 3840 = ₹240$$

$$\therefore \quad \text{Gain\%} = \frac{240}{3840} \times 100 = 6.25\%$$

Ex. 13 A vendor sells apples at 10 for ₹1 gaining 40%. How many apples did he buy for ₹1?

Sol. Given, SP of 10 apples = ₹1 and Gain = 40%

$$\text{CP of 10 apples} = \frac{100}{140} \times 1 = ₹\frac{5}{7}$$

$\because$ ₹$\frac{5}{7}$ yields 10 apples.

$\therefore$ ₹1 will yield $10 \times \frac{5}{7} = 14$ apples.

Ex. 14 A grocer buys 160 kg of rice at ₹27 per kg and mixes it with 240 kg of rice available at ₹32 per kg. At what rate per kg should he sell the mixture to gain 20% on the whole?

Sol. CP of 400 kg of rice = $160 \times 27 + 240 \times 32 = 4320 + 7680 = ₹12000$

Now, CP = ₹12000and Gain = 20%

$$\therefore \quad \text{SP} = \frac{120}{100} \times 12000 = ₹14400$$

$$\therefore \quad \text{SP per kg} = \frac{14400}{400} = ₹36$$

Ex. 15 A man purchases a certain number of apples at 3 per rupee and the same number of apples at 4 per rupee. He mixes them together and sells them at 3 per rupee. What is his gain or loss per cent?

Sol. CP of 1 apple for 1st rate $= ₹\frac{1}{3}$

CP of 1 apple for 2nd rate $= ₹\frac{1}{4}$

CP of 2 apples after mixing $= \frac{1}{3}+\frac{1}{4} = ₹\frac{7}{12}$

CP of 1 apple after mixing $= \frac{7}{12\times 2} = ₹\frac{7}{24}$

SP of 1 apple after mixing $= ₹\frac{1}{3}$

$\therefore$ Gain $= \frac{1}{3}-\frac{7}{24} = \frac{8-7}{24} = ₹\frac{1}{24}$

$\therefore$ Gain% $= \left(\frac{1}{24}\times\frac{24}{7}\times 100\right)\% = \frac{100}{7}\% = 14\frac{2}{7}\%$

11.1 Fast Track Techniques to solve the QUESTIONS

Technique 1

If a person sells two similar articles, one at a gain of a% and another at a loss of a%, then the seller always incurs a loss which is given by

$$\text{Loss\%} = \left(\frac{a}{10}\right)^2\%$$

Note: In this case, SP is immaterial.

Ex. 16 A man sold two radios for ₹2000 each. On one he gains 16% and on the other he losses 16%. Find his gain or loss per cent in the whole transaction.

Sol. Here, $a = 16\%$

According to the formula

$$\text{Loss\%} = \left(\frac{a}{10}\right)^2\% = \left(\frac{16}{10}\right)^2\%$$
$$= \frac{256}{100}\% = 2.56\%$$

Technique 2

If 'a'th part of some items is sold at x% loss, then required gain per cent in selling rest of the items in order that there is neither gain nor loss in whole transaction is

$$\frac{ax}{1-a}\%$$

Ex. 17 A medical store owner purchased medicines worth ₹6000 from a company. He sold 1/3 part of the medicine at 30% loss. On what gain he should sell his rest of the medicines, so that he has neither gain nor loss?

Sol. Given, $a = \frac{1}{3}$ and $x = 30\%$

According to the formula

$$\text{Required Gain\%} = \frac{ax}{1-a}\% = \frac{\frac{1}{3} \times 30}{1-\frac{1}{3}}$$
$$= \frac{10 \times 3}{2}\% = 15\%$$

Technique 3

A businessman sells his items at a Profit/Loss of a%. If he had sold it for ₹R more, he would have gained/lost b%. Then,

$$\text{CP of items } = \frac{R}{b \pm a} \times 100$$

Note:

'-' = When both are either profit or loss

'+' = When one is profit and other is loss

Ex. 18 A person sold a table at a profit of $6\frac{1}{2}$%. If he had sold it for ₹1250 more, he would have gained 19%. Find the CP of the table.

Sol. In this problem it is clearly stated that for ₹1250 more the gaint will rise to 19%. Hence

$$6\frac{1}{2}\% \text{ profit} + ₹1250 = 19\% \text{ profit}$$

$$\implies ₹1250 = \left(19 - \frac{13}{2}\right)\% \; CP$$

[Profit% is calculated on CP, hence we can write it % of CP]

$$\implies ₹1250 = \frac{38-13}{2}\% \; CP$$

$$\implies ₹1250 = \frac{25}{2} \times 100 \; CP$$

$\therefore$ CP $= 1250 \times 8 = ₹10000$

Fast Track Method

Here, $a = 6\frac{1}{2}\% = \frac{13}{2}$, $b = 19\%$ and $R = ₹1250$

According to the formula

$$\text{CP of table} = \frac{R}{b-a} \times 100$$
$$= \frac{1250}{19 - \frac{13}{2}} \times 100 = \frac{1250 \times 2}{25} \times 100 = ₹10000$$

Technique 4

If cost price of $'a'$ articles is equal to the selling price of $'b'$ articles, then

$$\text{Profit Percentage} = \frac{a-b}{b} \times 100\%$$

Ex. 19 If the cost price of 20 articles is equal to the selling price of 18 articles, then find the profit per cent.

Sol. Let CP of 1 article $= ₹x$

CP of 20 articles $= ₹20x =$ SP of 18 articles[Given]

$\therefore$ SP of 1 article $= \frac{20}{18}x = \frac{10x}{9}$

$$\text{Profit} = \frac{10x}{9} - x = \frac{x}{9}$$

$\therefore$ $$\text{Profit\%} = \frac{9}{x} \times 100 = \frac{100}{9} = 11\frac{1}{9}\%$$

Direct Approach

2 articles are gained on selling 18 articles

$\therefore$ $$\text{Profit\%} = \left(\frac{2}{18} \times 100\right)\% = 11\frac{1}{9}\%$$

Fast Track Method

Here, $a = 20$ and $b = 18$

According to the formula

$$\text{Profit\%} = \left(\frac{a-b}{b} \times 100\right)\% = \left(\frac{20-18}{18} \times 100\right)\%$$
$$= \frac{100}{9}\% = 11\frac{1}{9}\%$$

Technique 5

If a man purchases m items for ₹x and sells n items for ₹y, then

$$\text{Profit or Loss percent } = \frac{my - nx}{nx} \times 100\%$$

[Positive result means profit and negative result means loss]

Ex. 20 If Karan purchases 10 oranges for ₹25 and sells 9 oranges for ₹25, then find the gain percentage.

Sol. CP of 10 oranges = ₹25

Then, CP of 1 orange = ₹$\frac{25}{10}$

SP of 9 oranges = ₹25

SP of 1 orange = ₹$\frac{25}{9}$

$$\therefore \quad \text{Gain\%} = \frac{SP - CP}{CP} \times 100\%$$

$$= \frac{\frac{25}{9} - \frac{25}{10}}{\frac{25}{10}} \times 100\%$$

$$= \frac{25 \times 10 - 25 \times 9}{9 \times 10} \times \frac{10}{25} \times 100\%$$

$$= \frac{25}{9 \times 25} \times 100\% = \frac{100}{9}\% = 11\frac{1}{9}\%$$

Direct Approach

Let the total number of oranges be 90(LCM of 10 and 9).

Then, SP of 90 oranges $= 25 \times 10 =$ ₹ 250

CP of 90 oranges $= 25 \times 9 =$ ₹ 225

$$\therefore \quad \text{Profit percent} = \frac{SP - CP}{CP} \times 100\%$$

$$= \frac{250 - 225}{225} \times 100\%$$

$$= \frac{25}{225} \times 100\% = \frac{100}{9}\% = 11\frac{1}{9}\%$$

Fast Track Method

Here, $m = 20$, $x = 25$, $n = 9$ and $y = 25$

According to the formula

$$\text{Profit or Loss \%} = \frac{my - nx}{nx} \times 100\%$$
$$= \frac{25 \times 10 - 9 \times 25}{9 \times 25} \times 100\%$$
$$= \frac{250 - 225}{225} \times 100\%$$
$$= \frac{25}{225} \times 100\% = \frac{100}{9}\% = 11\frac{1}{9}\% \qquad \text{[positive sign indicates a profit]}$$

Ex. 21 Shakshi bought pens at a rate of 10 pens for ₹11 and sold them at a rate of 11 pens for ₹10. Then, find the profit or loss per cent.

Sol. CP of 10 pens = ₹11

Then, CP of 1 pen = $₹\frac{11}{10}$

Similarly, SP of 11 pens = ₹10

$\therefore$ SP of 1 pen = $₹\frac{10}{11}$

Here, SP < CP

$\therefore$ $\text{Loss Percent} = \frac{CP - SP}{CP} \times 100\% = \frac{\frac{11}{10} - \frac{10}{11}}{\frac{11}{10}} \times 100\%$

$$= \frac{121 - 100}{10 \times 11} \times \frac{10}{11} \times 100\%$$
$$= \frac{21}{121} \times 100\% = \frac{2100}{121}\% = 17\frac{43}{121}\%$$

Fast Track Method

Here, $m = 10$, $x = 11$, $n = 11$ and $y = 10$

According to the formula

$$\text{Profit or Loss percent} = \frac{my - nx}{nx} \times 100\%$$
$$= \frac{10 \times 10 - 11 \times 11}{11 \times 11} \times 100\%$$
$$= \frac{-21}{121} \times 100\% = -\frac{2100}{121}\%$$
$$= -17\frac{43}{121}\% \qquad \text{[negative sign indicates a loss]}$$

Technique 6

If A sold an article to B at a profit (loss) of r_1% and B sold this article to C at a profit (loss) of r_2%, then cost price of article for C is given by

$$\text{CP of article for C} = \text{CP of article for A} \times \left(1 \pm \frac{r_1}{100}\right)\left(1 \pm \frac{r_2}{100}\right)$$

[Positive for profit and negative for loss is used.] Ex.22 Nikunj sold a machine to Sonia at a profit of 30%. Sonia sold this machine to Anu at a loss of 20%. If Nikunj paid ₹5000 for this machine, then find the cost price of machine for Anu.

Sol. Nikunj sold the machine for a profit of 30

$$\therefore \text{ CP of machine for Sonia} = 130\% \text{ of CP of machine for Nikunj}$$

$$= \frac{130 \times 5000}{100} = ₹\ 6500$$

Now, Sonia sold the machine to Anu at a loss of 20%

$$\therefore \text{ CP of machine for Anu} = 80\% \text{ of CP of machine for Sonia}$$

$$= \frac{80 \times 6500}{100} = ₹\ 5200$$

Hence, CP of machine for Anu is ₹5200.

Fast Track Method

Here, $r_1 = 30\%$ and $r_2 = 20\%$

CP of machine for Nikunj = ₹ 5000

According to the formula

$$\therefore \text{ CP of machine for Anu} = \text{CP of machine for Nikunj} \times \left(1 + \frac{r_1}{100}\right)\left(1 - \frac{r_2}{100}\right)$$

$$= 5000\left(1 + \frac{30}{100}\right)\left(1 - \frac{20}{100}\right)$$

$$= 5000 \times \frac{130}{100} \times \frac{80}{100} = ₹\ 5200$$

Technique 7

If a dishonest trader professes to sell his items at CP but uses false weight, then

$$\text{Gain\%} = \frac{\text{Error}}{\text{True value - Error}} \times 100\%$$

$$\text{Gain\%} = \frac{\text{True Weight - False Weight}}{\text{False Weight}} \times 100\%$$

Here, while calculating gain or profit per cent, we have taken false weight as a base. Because CP is what is paid when an item is purchased or manufactured. Here, in this case dishonest trader is telling false weight to be the CP and he is gaining only when sells at false weight.

Ex.23 A dishonest dealer professes to sell his goods at cost price but he uses a weight of 930 g for 1 kg weight. Find his gain per cent.

Sol. 70g is gained on 930 g.

$$\therefore \quad \text{Gain\%} = \left(\frac{70}{930} \times 100\right)\% = 7\frac{49}{93}\%$$

Fast Track Method

According to the formula,

$$\begin{aligned}\text{Gain\%} &= \frac{\text{Error}}{\text{True value - Error}} \times 100\% \\ &= \frac{700}{1000-70} \times 100\% \\ &= \frac{700}{930} \times 100\% = \frac{700}{93}\% = 7\frac{49}{93}\%\end{aligned}$$

Technique 8

If a shopkeeper sells his goods at a% loss on cost price but uses b g instead of c g, then

$$\text{Percentage profit or loss} = \left[(100-a)\frac{c}{b} - 100\right]\%$$

[Sign is Positive for Profit and Negative for Loss]

Ex. 24 A dealer sells goods at 6% loss on cost price but uses 14 g instead of 16 g. What is his percentage profit or loss?

Sol. Given, $a = 6\%, b = 14\ g$ and $c = 16g$

According to the formula,

$$\begin{aligned}\text{Required answer} &= \left[(100-a)\frac{c}{b} - 100\right]\% \\ &= \left[(100-6)\frac{16}{14} - 100\right]\% = \left[(94)\frac{8}{7} - 100\right]\% \\ &= \left(\frac{752-700}{7}\right)\% = \frac{52}{7}\% \\ &= 7\frac{3}{7}\% \text{ gain} \qquad \text{[+ive sign shows that there is a gain]}\end{aligned}$$

Technique 9

If a dealer sells his goods at a% profit or loss on cost price and uses b% less weight, then

$$\text{Percentage profit or loss} = \frac{(b \pm a)}{100 - b} \times 100\%$$

[Sign is Positive for Profit and Negative for Loss]

Ex. 25 A dealer sells his goods at 20% loss on cost price but uses 40% less weight. What is his percentage profit or loss?

Sol. Given, $a = 20\%$ and $b = 40\%$

According to the formula,

$$\text{Required answer} = \frac{(b \pm a)}{100 - b} \times 100\% = \frac{(40 - 20)}{100 - 40} \times 100\%$$
$$= \frac{2}{6} \times 100\% = \frac{100}{3}\% = 33\frac{1}{3}\% \text{ profit}$$

Technique 10

If 'a' part of an article is sold at x% profit/loss, 'b' part at y% profit/loss and 'c' part at z% profit/loss and finally there is a profit/loss of R, then

$$\text{Cost price of entire article} = ₹\ \frac{R \times 100}{ax + by + cz}$$

Ex. 26 If $\frac{2}{3}$ part of an article is sold at 30% profit, $\frac{1}{4}$ part at 16% profit and remaining part at 12% profit and finally, there is a profit of ₹75, then find the cost price of the article

Sol. Here, $a = \frac{2}{3}$, $x = 30\%$, $b = \frac{1}{4}$, $y = 16\%$, $z = 12\%$ and $R = ₹75$

Then, remaining part $c = 1 - \left(\frac{2}{3} + \frac{1}{4}\right) = \frac{1}{12}$

$\therefore$ Cost price of entire article $= \frac{R \times 100}{ax + by + cz}$

$$= \frac{75 \times 100}{\frac{2}{3} \times 30 + \frac{1}{4} \times 16 + \frac{1}{12} \times 12}$$
$$= \frac{75 \times 100}{20 + 4 + 1} = \frac{7500}{25} = ₹300$$